

5-4 Area of Regular Polygons

Architecture Firm | Standard: 6.G.1 | Color-by-Number Practice

Learning Goal: I can find the area of a regular polygon by decomposing it into triangles.

Directions: Solve each problem. Match each answer to the Color Code. On the next page, color every box with that problem number using the matching color.

Vocabulary: perimeter, apothem, regular polygon

Explain one: I solved problem ___ by ___. Because ___, my answer is ___.

Solve First

1. A regular polygon has perimeter 6 and apothem 4. Find its area.

Work: _____

Answer: _____

2. A regular polygon has perimeter 8 and apothem 5. Find its area.

Work: _____

Answer: _____

3. A regular polygon has perimeter 10 and apothem 6. Find its area.

Work: _____

Answer: _____

4. A regular polygon has perimeter 12 and apothem 7. Find its area.

Work: _____

Answer: _____

5. A regular polygon has perimeter 5 and apothem 8. Find its area.

Work: _____

Answer: _____

6. A regular polygon has perimeter 7 and apothem 10. Find its area.

Work: _____

Answer: _____

7. A regular polygon has perimeter 9 and apothem 12. Find its area.

Work: _____

Answer: _____

8. A regular polygon has perimeter 11 and apothem 6. Find its area.

Work: _____

Answer: _____

9. A regular polygon has perimeter 13 and apothem 4. Find its area.

Work: _____

Answer: _____

10. A regular polygon has perimeter 14 and apothem 5. Find its area.

Work: _____

Answer: _____

11. A regular polygon has perimeter 15 and apothem 8. Find its area.

Work: _____

Answer: _____

12. A regular polygon has perimeter 16 and apothem 3. Find its area.

Work: _____

Answer: _____

Quick check: Before coloring, make sure every answer appears in the Color Code on the next page.

5-4 Mystery Picture: Blueprint Mosaic

Solve a problem, match the answer to a color, then color every box with that problem number.

Color Code	Color Code	Color Code	Color Code
12 = red	20 = orange	30 = yellow	42 = green
35 = blue	54 = purple	33 = pink	26 = brown
60 = gray	24 = teal		

2	2	1	2	2	2	3	3	3	4	4	6
2	5	5	5	5	5	6	6	6	7	9	6
1	5	8	12	8	8	9	9	9	12	9	6
3	6	10	2	2	1	2	2	4	2	11	6
3	6	9	2	5	5	5	7	4	2	9	6
3	6	9	11	5	8	11	7	4	12	9	6
5	8	11	3	6	12	1	10	4	2	11	7
5	8	11	3	8	6	10	8	5	2	11	8
5	8	11	5	5	5	4	5	5	3	11	8
7	10	3	4	4	2	3	3	4	5	2	10
7	12	12	1	11	11	12	12	12	2	4	11
9	9	9	8	8	8	4	9	9	10	11	11

Early finisher writing: Choose one problem and explain your thinking in one complete sentence: Because ___, I ___.

Teacher Answer Key

#	Answer	Color	Teacher Tip
1	12	red	$A = 1/2 \times \text{perimeter} \times \text{apothem}.$
2	20	orange	$A = 1/2 \times \text{perimeter} \times \text{apothem}.$
3	30	yellow	$A = 1/2 \times \text{perimeter} \times \text{apothem}.$
4	42	green	$A = 1/2 \times \text{perimeter} \times \text{apothem}.$
5	20	orange	$A = 1/2 \times \text{perimeter} \times \text{apothem}.$
6	35	blue	$A = 1/2 \times \text{perimeter} \times \text{apothem}.$
7	54	purple	$A = 1/2 \times \text{perimeter} \times \text{apothem}.$
8	33	pink	$A = 1/2 \times \text{perimeter} \times \text{apothem}.$
9	26	brown	$A = 1/2 \times \text{perimeter} \times \text{apothem}.$
10	35	blue	$A = 1/2 \times \text{perimeter} \times \text{apothem}.$
11	60	gray	$A = 1/2 \times \text{perimeter} \times \text{apothem}.$
12	24	teal	$A = 1/2 \times \text{perimeter} \times \text{apothem}.$

Use note: This file is original practice aligned to the lesson skill. It is designed for low-prep review, centers, sub plans, or early finishers.